

Multivariate Bayesian P-spline estimation of spectral density matrices, with application to LISA TDI noise

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We present a Bayesian P-spline method for estimating the frequency-dependent cross-spectral density matrix of stationary multivariate time series. The inverse spectral matrix is parametrised through its frequency-varying Cholesky decomposition, which guarantees Hermitian positive definiteness at every frequency. Each real log-diagonal entry and each real and imaginary off-diagonal entry is given an independent penalised B-spline prior that controls smoothness. Inference uses a blocked, coarse-grained Whittle likelihood with safe-Bayes η -tempering to stabilise posterior calibration, sampled by the No-U-Turn Sampler from a variational initialisation. On synthetic VAR(2) benchmarks with known ground truth, the method recovers both diagonal and cross-spectral structure, attains near-nominal credible-interval coverage, and achieves a relative integrated squared (Frobenius) error (RISE) that decreases with sample size. We then apply the method to publicly released simulated LISA time-delay interferometry (TDI) data in two noise configurations. In the idealised symmetric case, the full multivariate model and a reduced model that assumes a diagonal AET noise covariance agree to within $\sim 10^{-3}$ in RISE. Under realistic noise that is asymmetric across the six Movable Optical Sub-Assemblies (MOSAs), the AET-diagonal assumption fails by more than an order of magnitude in RISE ($\sim 3.3 \times 10^{-2}$ versus $\sim 10^{-3}$), whereas the full multivariate model recovers the cross-spectral structure.

I. INTRODUCTION

Accurate estimation of the noise power spectral density (PSD) underpins gravitational-wave (GW) data analysis: search, parameter estimation, and stochastic-background inference all require a reliable noise model across the detector band. For LIGO and Virgo, parametric models augmented with spline components, most notably BayesLine [1], are widely used. The space-based detector LISA [2] poses a harder problem: its three time-delay interferometry (TDI) channels are correlated, and their noise properties vary smoothly but non-trivially across the millihertz band. A single-channel noise model cannot capture the cross-spectral correlations that matter for joint inference.

Flexible nonparametric Bayesian approaches to PSD estimation have a well-developed statistical foundation, characterised by an expansion of the PSD in terms of a potentially large or infinite number of basis functions combined with noninformative priors on the coefficients. Examples include Bernstein polynomials with weights induced by Dirichlet process priors [3], cosines with a discounted regularized horseshoe prior [4], and B-splines with knots and weights induced by Dirichlet process priors [5]. With an a priori unknown and potentially infinite number of basis functions, posterior computation requires trans-dimensional Markov chain Monte Carlo (MCMC) algorithms, which are computationally expensive. Penalised B-splines (P-splines) [6] avoid the trans-dimensional step by representing smooth functions as linear combinations of a finite but large number of B-spline basis functions, with a roughness penalty on adjacent coefficients that prevents overfitting. In a Bayesian

formulation this penalty can be incorporated naturally into the prior distribution of the spline coefficients. Applied to spectral density estimation via the Whittle likelihood, this yields a smooth, data-driven estimator that requires no parametric noise template. Univariate Bayesian P-spline PSD methods were introduced by Maturana-Russel and Meyer [7]. They have since been applied to GW noise estimation, including recent LISA-specific work that uses log-spline representations tuned to the steep low-frequency rise of LISA noise [8]. Related non-parametric strategies model unknown noise shapes in stochastic-background searches with other spline families rather than penalised B-splines. These include cubic B-spline single-link noise models analysed by Bayesian model selection [9], and Akima-spline noise components introduced at the TDI level with the number and location of knots inferred by a reversible-jump MCMC [10].

These univariate approaches model each channel independently. This is justified in the idealised equal-arm stationary LISA configuration, which has vanishing off-diagonal cross-spectra between the three TDI channels, but realistic noise — with unequal arm lengths, noise-level imbalance across the six Movable Optical Sub-Assemblies (MOSAs), and instrumental breathing [11] — generates non-trivial off-diagonal entries in the 3×3 spectral density matrix $\mathbf{S}(f)$, whose normalised magnitudes (the channel-pair coherences) quantify the cross-channel correlation that single-channel models discard. Treating these cross-spectra as zero when they are not can degrade sensitivity to stochastic GW backgrounds and bias parameter estimation [12, 13]. Any extension of univariate methods to nonparametrically model $\mathbf{S}(f)$ needs to ensure Hermitian positive definiteness at every frequency

while allowing each matrix element to vary smoothly with frequency.

Existing nonparametric treatments of LISA TDI spectral matrices follow two broad routes. Muratore *et al.* [14] model the diagonal PSDs and the real and imaginary parts of the cross-spectra of $\mathbf{S}(f)$ as smooth fractional deviations from a design spectrum, each represented by a natural cubic spline. They note that this construction is not guaranteed to yield a positive-definite matrix away from the design point, and that it is well suited to their local Fisher-matrix study of noise-knowledge uncertainty rather than to fitting data directly. A second route models the single-link noise log-PSD nonparametrically with cubic B-splines and maps it to the TDI variables through the known TDI transfer functions [9]. Because the resulting TDI covariance is built from a valid single-link covariance through this linear transfer, it is Hermitian positive definite by construction, at the cost of assuming the simplified single-link noise structure.

In this paper, we model the PSD matrix of the XYZ TDI variables directly by extending the Bayesian P-spline PSD framework [7] to the multivariate p -dimensional setting. We parametrise $\mathbf{S}(f)^{-1}$ through its Cholesky decomposition [15, 16], which factorises the posterior density into p independent components that can be sampled in parallel and guarantees the Hermitian positive definiteness condition. Each block’s log-diagonal spectra and real and imaginary parts of the complex off-diagonal cross-spectra are modelled with penalised B-splines under a hierarchical smoothing prior. This prior is updated by a blocked, coarse-grained Whittle likelihood with safe-Bayes η -tempering [17] to stabilise posterior calibration. Each chain is initialised from a draw of a stochastic variational approximation [18], with sampling performed by the No-U-Turn Sampler [19]. We validate the method on a three-channel VAR(2) simulation with known ground truth, and on publicly released simulated LISA TDI datasets in a symmetric and a physically realistic asymmetric per-MOSA noise configuration. In the asymmetric case an AET-restricted model fails by more than an order of magnitude in relative integrated squared (Frobenius) error while the full multivariate model recovers the cross-spectral structure. A further benchmark against the bivariate estimators of Liu *et al.* [20] on a VAR(2) problem with closed-form ground truth is reported in Appendix B.

The remainder of the paper is organised as follows. Section II develops the statistical model and the inference scheme. Section III presents simulation studies on a 3D VAR(2) benchmark and on publicly released simulated LISA TDI data. Section IV summarises the results, discusses limitations and future directions.

II. MULTIVARIATE P-SPLINES

A. Likelihood

A glossary of the symbols used throughout this paper is provided in Table III (Appendix A). Readers may find it useful to keep this table open while reading the rest of this section. Let $\mathbf{Z} = (\mathbf{Z}_1, \dots, \mathbf{Z}_n)^\top \in \mathbb{R}^{n \times p}$ be a p -dimensional stationary, mean-zero time series sampled at intervals $\Delta_t = 1/(2f_{Ny})$, so that $\mathbf{Z}_t = \mathbf{Z}(t\Delta_t)$ for $t = 1, \dots, n$, where f_{Ny} is the Nyquist frequency. The total observation time is $T = n\Delta_t$ and the frequency resolution is

$$\Delta_f = \frac{1}{n\Delta_t} = \frac{1}{T}. \quad (1)$$

The discrete Fourier transform (DFT) of $\mathbf{Z}$ is

$$\mathbf{d}(f_k) = \Delta_t \sum_{t=1}^n \mathbf{Z}_t \exp\left(-2\pi i \frac{k}{n} t\right), \quad (2)$$

with $f_k = k/T$ for $k = 1, \dots, N$, where $N = \lfloor n/2 \rfloor$.

For stationary series with absolutely summable matrix autocovariances $\sum_{\tau=-\infty}^{\infty} \|\Gamma(\tau)\| < \infty$, the Fourier coefficients $\mathbf{d}(f_k)$ are asymptotically independent, complex Gaussian with mean zero and covariance $T \mathbf{S}(f_k)$, where

$$\mathbf{S}(f) = \frac{1}{2f_{Ny}} \sum_{\tau=-\infty}^{\infty} \Gamma(\tau) \exp(-2\pi i f \tau \Delta_t)$$

is the two-sided spectral density matrix — the Fourier transform of the autocovariance $\Gamma(\tau) = \mathbb{E}(\mathbf{Z}_t \mathbf{Z}_{t+\tau}^\top)$ at integer lag τ . While the likelihood below is expressed in terms of the two-sided density, all spectra reported in this paper (estimates, analytic references, and figures) follow the one-sided convention, obtained by doubling the two-sided density at positive frequencies. This asymptotic Gaussian approximation is the basis of the multivariate Whittle likelihood [21]

$$\mathcal{L}(\mathbf{d}|\mathbf{S}) \propto \prod_{k=1}^N |\mathbf{S}(f_k)|^{-1} \times \exp\left(-\frac{1}{T} \mathbf{d}(f_k)^* \mathbf{S}(f_k)^{-1} \mathbf{d}(f_k)\right), \quad (3)$$

where $\mathbf{d}(f_k)^*$ denotes the conjugate transpose of $\mathbf{d}(f_k)$ and $|\cdot|$ denotes the matrix determinant. In general a spectral density matrix is only guaranteed to be Hermitian positive semidefinite. The Whittle likelihood, however, requires $\mathbf{S}(f_k)^{-1}$ to exist, so we assume strict positive definiteness throughout (a property the Cholesky parametrisation of Section II B enforces by construction).

The diagonal entries $S_{ii}(f)$ are the power spectral densities (PSDs) of channel i , while the off-diagonals $S_{ij}(f)$ for $i \neq j$ are the (complex) cross-spectra. A scale-free

summary of the linear dependence between channels i and j at frequency f is the complex coherence,

$$C_{ij}(f) = \frac{S_{ij}(f)}{\sqrt{S_{ii}(f)S_{jj}(f)}}, \quad |C_{ij}(f)| \in [0, 1], \quad \text{•} \quad (4)$$

whose magnitude is zero when channels i and j are uncorrelated at frequency f and unity when they are perfectly linearly related. We use $|C_{ij}(f)|$ to visualise off-diagonal recovery in the LISA application (Sec. III B).

Blocking. To trade frequency resolution for periodogram stability we partition $\mathbf{Z}$ into N_b equal, non-overlapping blocks $(\mathbf{Z}^{(1)}, \dots, \mathbf{Z}^{(N_b)})$, each of duration $T_b = T/N_b$ and containing n/N_b samples (illustrated in Fig. 1). Each block may be tapered by a window function w_t before the DFT to suppress spectral leakage. Writing the number of samples per block as $n_b = n/N_b$, we define the window's equivalent noise bandwidth as

$$\text{ENBW} = n_b \frac{\sum_{t=1}^{n_b} w_t^2}{(\sum_{t=1}^{n_b} w_t)^2}. \quad \text{•} \quad (5)$$

We rescale each block periodogram by $1/\text{ENBW}$, so that the complex-Wishart mean of $\mathbf{Y}(f_k)$ remains $T_b \mathbf{S}(f_k)$. The separate effect of tapering on the effective likelihood information is described below. Denoting the DFT of block i by $\mathbf{d}^{(i)}$, and assuming blocks are approximately independent under stationarity, the blocked Whittle likelihood factorises as

$$\mathcal{L}_b(\mathbf{d}|\mathbf{S}) = \prod_{i=1}^{N_b} \mathcal{L}(\mathbf{d}^{(i)}|\mathbf{S}). \quad (6)$$

Collecting the sum over blocks into a single Hermitian matrix

$$\mathbf{Y}(f_k) = \sum_{i=1}^{N_b} \mathbf{I}^{(i)}(f_k) = N_b \bar{\mathbf{I}}(f_k), \quad (7)$$

where $\mathbf{I}^{(i)}(f_k) = \mathbf{d}^{(i)}(f_k) \mathbf{d}^{(i)}(f_k)^*$ is the block- i periodogram and $\bar{\mathbf{I}}(f) = N_b^{-1} \sum_{i=1}^{N_b} \mathbf{I}^{(i)}(f)$ is the block-averaged periodogram (the Welch estimator), and applying the trace cyclic property $\text{tr}(ABC) = \text{tr}(CAB)$, the blocked likelihood reduces to

$$\mathcal{L}_b(\mathbf{d}|\mathbf{S}) \propto \prod_{k=1}^{n/(2N_b)} |\mathbf{S}(f_k)|^{-N_b} \times \exp\left(-\frac{1}{T_b} \text{tr}[\mathbf{S}(f_k)^{-1} \mathbf{Y}(f_k)]\right). \quad (8)$$

Under the asymptotic Gaussian approximation, $\mathbf{Y}(f_k)$ follows a complex Wishart distribution¹ from summing

N_b assumed-independent single-block periodogram contributions:

$$\mathbf{I}^{(i)}(f_k) \sim \mathcal{CW}_p(T_b \mathbf{S}(f_k), 1), \quad (9)$$

$$\mathbf{Y}(f_k) \sim \mathcal{CW}_p(T_b \mathbf{S}(f_k), N_b). \quad (10)$$

Eigendecomposition. Because $\mathbf{Y}(f_k)$ is Hermitian positive semidefinite, it admits an eigendecomposition $\mathbf{Y}(f_k) = \sum_{\nu=1}^p \lambda_{\nu}^{(k)} \mathbf{v}_{\nu}^{(k)} \mathbf{v}_{\nu}^{(k)*} = \sum_{\nu=1}^p \mathbf{u}_{\nu}^{(k)} \mathbf{u}_{\nu}^{(k)*}$, with rescaled eigenvectors $\mathbf{u}_{\nu}^{(k)} = \sqrt{\lambda_{\nu}^{(k)}} \mathbf{v}_{\nu}^{(k)}$. Substituting into the trace in (8) gives the eigenvector form of the blocked likelihood,

$$\mathcal{L}_b(\mathbf{d}|\mathbf{S}) \propto \prod_{k=1}^{n/(2N_b)} |\mathbf{S}(f_k)|^{-N_b} \times \exp\left(-\frac{1}{T_b} \sum_{\nu=1}^p \mathbf{u}_{\nu}^{(k)*} \mathbf{S}(f_k)^{-1} \mathbf{u}_{\nu}^{(k)}\right). \quad (11)$$

This follows from the spectral decomposition of the Hermitian positive semidefinite $\mathbf{Y}(f_k)$ and the linearity of the trace. The eigenvector representation is used computationally because it turns the trace contribution into a sum of quadratic forms in the Cholesky parameters, which is the form that factorises into the per-channel regressions below.

Coarse graining. Blocking reduces the number of periodogram matrices from N to $n/(2N_b)$, but each of the remaining frequencies still requires a separate evaluation of $\mathbf{S}(f_k)^{-1}$. Where $\mathbf{S}(f)$ is slowly varying, we can pool adjacent frequencies to gain further speed (illustrated in Fig. 1). Partition the block frequencies into N_c disjoint subsets $\{f_1, \dots, f_{n/(2N_b)}\} = \bigcup_{h=1}^{N_c} J_h$, each containing N_h adjacent frequencies with bin centre $\bar{f}_h \equiv \frac{1}{N_h} \sum_{f \in J_h} f$, which need not coincide with a Fourier frequency. Replace the per-frequency matrix $\mathbf{Y}(f_k)$ by the within-bin sum $\mathbf{Y}_h^{\text{cg}} = \sum_{f \in J_h} \mathbf{Y}(f)$.

Under the assumption that $\mathbf{S}(f) \approx \mathbf{S}(\bar{f}_h)$ across each J_h , the $\mathbf{Y}(f)$ within the bin are approximately i.i.d. $\mathcal{CW}_p(T_b \mathbf{S}(\bar{f}_h), N_b)$, and by the additivity of the complex Wishart under a common scale matrix,

$$\mathbf{Y}_h^{\text{cg}} \sim \mathcal{CW}_p(T_b \mathbf{S}(\bar{f}_h), N_b N_h). \quad \text{•} \quad (12)$$

The approximation is only as good as the constant- $\mathbf{S}$ assumption, so bin edges must avoid sharp spectral features (peaks, lines, transfer-function nulls).

The coarse-grained likelihood then takes the same form as Eq. (11) with the $n/(2N_b)$ block frequencies replaced by N_c coarse bins and a Wishart shape parameter $N_b N_h$ (the number of fine periodograms summed per coarse bin):

$$\mathcal{L}_{b,c}(\mathbf{d}|\mathbf{S}) \propto \prod_{h=1}^{N_c} |\mathbf{S}(\bar{f}_h)|^{-N_b N_h} \times \exp\left(-\frac{1}{T_b} \text{tr}[\mathbf{S}(\bar{f}_h)^{-1} \mathbf{Y}_h^{\text{cg}}]\right), \quad \text{•} \quad (13)$$

¹ We write $\mathbf{A} \sim \mathcal{CW}_d(\mathbf{M}, m)$ if the $d \times d$ Hermitian positive definite matrix $\mathbf{A}$ has density proportional to $|\mathbf{M}|^{-m} \exp(-\text{tr}[\mathbf{M}^{-1} \mathbf{A}])$, with mean $m\mathbf{M}$.

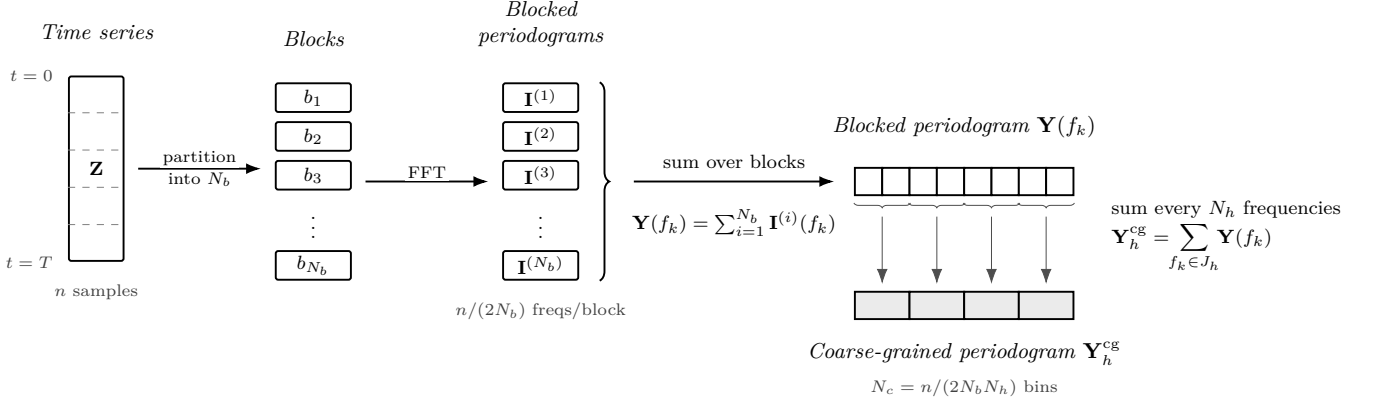


FIG. 1. Construction of the blocked, coarse-grained periodogram. The n -sample time series is partitioned into N_b non-overlapping blocks $\{b_i\}$ of length T/N_b . Each block is Fourier-transformed independently, yielding N_b single-block periodograms $\mathbf{I}^{(i)}$ each of length $n/(2N_b)$. These are summed across blocks to form the blocked periodogram $\mathbf{Y}(f_k) = \sum_{i=1}^{N_b} \mathbf{I}^{(i)}(f_k)$ (related to the Welch estimator by $\bar{\mathbf{I}} = \mathbf{Y}/N_b$), which is then summed over consecutive groups of N_h adjacent fine frequencies to produce the final coarse-grained periodogram $\mathbf{Y}_h^{\text{cg}}$ of length $n/(2N_b N_h)$.

with eigendecomposition

$$\mathbf{Y}_h^{\text{cg}} = \sum_{\nu=1}^p \mathbf{u}_{\nu}^{(h)} \mathbf{u}_{\nu}^{(h)*}, \quad (14)$$

$$\mathbf{u}_{\nu}^{(h)} = \sqrt{\lambda_{\nu}^{(h)}} \mathbf{v}_{\nu}^{(h)}. \quad (15)$$

Although rescaling the periodogram by $1/\text{ENBW}$ preserves its mean, tapering correlates neighbouring Fourier coefficients and therefore reduces the effective number of independent frequency-domain observations. We approximate this loss of information by dividing the log-likelihood by the window's equivalent noise bandwidth in Eq. (5), or equivalently by using the power likelihood

$$\mathcal{L}_{b,c,w}(\mathbf{d} | \mathbf{S}) \propto \mathcal{L}_{b,c}(\mathbf{d} | \mathbf{S})^{1/\text{ENBW}}. \quad (16)$$

Thus a rectangular window leaves the likelihood unchanged, whereas a non-rectangular taper reduces its curvature by the factor $1/\text{ENBW}$.

B. Parametrisation using the Cholesky Decomposition

Following Rosen and Stoffer [15], Hu and Prado [16], we parametrise the *inverse* spectral density matrix through its Cholesky decomposition. Working with $\mathbf{S}^{-1}$ rather than $\mathbf{S}$ is what makes the problem tractable: the

quadratic form $\mathbf{u}^* \mathbf{S}^{-1} \mathbf{u}$ splits into a sum of p decoupled quadratic terms, which in turn factorises the likelihood into p independent regressions (one per channel) that can be sampled in parallel. We write

$$\mathbf{S}(f_k)^{-1} = \mathbf{T}_k^* \mathbf{D}_k^{-1} \mathbf{T}_k, \quad (17)$$

where $\mathbf{D}_k$ is diagonal with positive entries $\delta_{1k}^2, \delta_{2k}^2, \dots, \delta_{pk}^2$, and

$$\mathbf{T}_k = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\theta_{21}^{(k)} & 1 & 0 & \cdots & 0 \\ -\theta_{31}^{(k)} & -\theta_{32}^{(k)} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -\theta_{p1}^{(k)} & -\theta_{p2}^{(k)} & \cdots & -\theta_{p,p-1}^{(k)} & 1 \end{pmatrix} \quad (18)$$

is a $p \times p$ complex unit lower triangular matrix with $\theta_{il}^{(k)}$ representing the complex-valued off-diagonal elements for $i > l$.

This parametrisation factorises the likelihood into a product of p per-channel terms,

$$\mathcal{L}(\mathbf{d} | \mathbf{S}) \propto \prod_{j=1}^p \mathcal{L}_j(\boldsymbol{\theta}_j, \boldsymbol{\delta}_j), \quad (19)$$

one for each row of $\mathbf{T}$. Each $\mathcal{L}_j$ is a univariate Whittle regression of the j th channel of the eigenvector pseudo-observations onto the preceding $(j-1)$ channels:

$$\mathcal{L}_j(\boldsymbol{\theta}_j, \boldsymbol{\delta}_j) \propto \prod_{h=1}^{N_c} \delta_{jh}^{-2N_b N_h} \exp \left(\frac{-\sum_{\nu=1}^p \left| u_{j\nu}^{(h)} - \sum_{l=1}^{j-1} \theta_{jl}^{(h)} u_{l\nu}^{(h)} \right|^2}{T_b \delta_{jh}^2} \right), \quad (20)$$

with regression coefficients $\theta_{jl}^{(h)}$ (the off-diagonal Cholesky entries at coarse bin h) and noise variance δ_{jh}^2 (the j th diagonal entry of $\mathbf{D}_h$). The eigenvectors $\mathbf{u}_\nu^{(h)}$ are those of the coarse-grained statistic $\mathbf{Y}_h^{\text{cg}}$, evaluated

at the bin midpoint $\bar{f}_h$. The parameter blocks $\boldsymbol{\theta}_j$ and $\boldsymbol{\delta}_j$ collect all $\theta_{jl}^{(h)}$ ($l < j$) and δ_{jh} across the coarse bins $h = 1, \dots, N_c$.

In particular, for $p = 3$ channels (as in the LISA case), the three factors depend on the parameter subvectors $\boldsymbol{\delta}_1$, $(\boldsymbol{\delta}_2, \boldsymbol{\theta}_2)$, and $(\boldsymbol{\delta}_3, \boldsymbol{\theta}_3)$, respectively, as follows:

$$\mathcal{L}_1(\boldsymbol{\delta}_1) \propto \prod_{h=1}^{N_c} \delta_{1h}^{-2N_b N_h} \exp \left(\frac{-\sum_{\nu=1}^p |u_{1\nu}^{(h)}|^2}{T_b \delta_{1h}^2} \right), \quad (21)$$

$$\mathcal{L}_2(\boldsymbol{\theta}_2, \boldsymbol{\delta}_2) \propto \prod_{h=1}^{N_c} \delta_{2h}^{-2N_b N_h} \exp \left(\frac{-\sum_{\nu=1}^p |u_{2\nu}^{(h)} - \theta_{21}^{(h)} u_{1\nu}^{(h)}|^2}{T_b \delta_{2h}^2} \right), \quad (22)$$

$$\mathcal{L}_3(\boldsymbol{\theta}_3, \boldsymbol{\delta}_3) \propto \prod_{h=1}^{N_c} \delta_{3h}^{-2N_b N_h} \exp \left(\frac{-\sum_{\nu=1}^p |u_{3\nu}^{(h)} - \theta_{31}^{(h)} u_{1\nu}^{(h)} - \theta_{32}^{(h)} u_{2\nu}^{(h)}|^2}{T_b \delta_{3h}^2} \right). \quad (23)$$

Note that a factorised likelihood does not in general imply a factorised posterior. Here, however, the posterior also factorises because we place independent priors across these three parameter blocks,

$$\pi(\boldsymbol{\delta}_1, \boldsymbol{\delta}_2, \boldsymbol{\theta}_2, \boldsymbol{\delta}_3, \boldsymbol{\theta}_3) = \pi_1(\boldsymbol{\delta}_1) \pi_2(\boldsymbol{\delta}_2, \boldsymbol{\theta}_2) \pi_3(\boldsymbol{\delta}_3, \boldsymbol{\theta}_3), \quad (24)$$

$$p(\boldsymbol{\delta}_1, \boldsymbol{\delta}_2, \boldsymbol{\theta}_2, \boldsymbol{\delta}_3, \boldsymbol{\theta}_3 | \mathbf{d}) \propto \mathcal{L}_1(\boldsymbol{\delta}_1) \pi_1(\boldsymbol{\delta}_1) \times \mathcal{L}_2(\boldsymbol{\delta}_2, \boldsymbol{\theta}_2) \pi_2(\boldsymbol{\delta}_2, \boldsymbol{\theta}_2) \times \mathcal{L}_3(\boldsymbol{\delta}_3, \boldsymbol{\theta}_3) \pi_3(\boldsymbol{\delta}_3, \boldsymbol{\theta}_3). \quad (25)$$

Hence Bayes' rule yields three independent posterior factors corresponding to $\boldsymbol{\delta}_1$, $(\boldsymbol{\delta}_2, \boldsymbol{\theta}_2)$, and $(\boldsymbol{\delta}_3, \boldsymbol{\theta}_3)$. This means we can run an MCMC chain for each block separately and in parallel, which in general reduces the computation time.

C. Safe-Bayes η -tempering

The Whittle likelihood is an asymptotic approximation to the exact Gaussian likelihood of $\mathbf{Z}$, and the coarse-grained form in Eq. (13) compounds this with a within-bin constant- $\mathbf{S}$ assumption. Both approximations introduce mild misspecification. A well-known consequence of likelihood misspecification is that the posterior can over-concentrate, producing credible intervals that under-cover the truth even when the posterior mode is accurate [17].

We mitigate this with the safe-Bayes correction of Grünwald and van Ommen [17]: raise the likelihood to a

power $\eta \in (0, 1]$,

$$\mathcal{L}_\eta(\mathbf{d} | \mathbf{S}) \propto \mathcal{L}_{b,c,w}(\mathbf{d} | \mathbf{S})^\eta \propto \mathcal{L}_{b,c}(\mathbf{d} | \mathbf{S})^{\eta/\text{ENBW}}, \quad (26)$$

which, together with the ENBW correction, gives an effective scaling $N_b N_h \mapsto \eta N_b N_h / \text{ENBW}$ in Eq. (13). The limit $\eta \rightarrow 1$ recovers the window-corrected likelihood, while $\eta \rightarrow 0$ recovers the prior. Intermediate η down-weights the (asymptotic) likelihood relative to the prior in exactly the right amount to restore nominal coverage when the approximation is imperfect.

A useful way to read this correction is through the curvature of the log-likelihood. From Eq. (13), the observed Fisher information per coarse bin scales linearly with the number of fine periodograms summed in each bin: schematically,

$$-\partial_{\theta\theta}^2 \log \mathcal{L}_{b,c,w}(\mathbf{d} | \mathbf{S}(\theta)) \propto \frac{N_b N_h}{\text{ENBW}} \mathcal{I}_0(\theta), \quad (27)$$

where $\mathcal{I}_0$ is the per-bin information matrix induced by a single complex Wishart observation. For a fixed window, increasing either blocking or coarse-graining therefore sharpens the log-posterior by the same factor $N_b N_h$, while tapering offsets this by $1/\text{ENBW}$. We treat η as a tunable calibration knob rather than a fundamental quantity. For the small synthetic benchmarks in Section III A ($n = 16,384$, $N_b N_h \lesssim 10^2$) the un-tempered limit $\eta = 1$ is already well-calibrated and is used throughout. The correction becomes important for the LISA application (Section III B), where $N_b N_h$ is two to three orders of magnitude larger and the resulting log-likelihood curvature drives both posterior over-concentration and, at sufficiently large $N_b N_h$, sampler pathologies (maximum tree-depth saturation in NUTS). An η -sensitivity

study reported in Appendix C motivates the choice of η used for the LISA results.

D. P-spline modelling

We model the frequency-dependent Cholesky parameters with penalised B-splines,

$$\log \delta_{jk}^2 = \sum_{m=1}^{K_j} B_m(f_k) w_{j,m}^{(\delta)} \quad (28)$$

$$\Re[\theta_{jl}(f_k)] = \sum_{m=1}^{K_{jl}} B_m(f_k) w_{jl,m}^{(\Re)} \quad (29)$$

$$\Im[\theta_{jl}(f_k)] = \sum_{m=1}^{K_{jl}} B_m(f_k) w_{jl,m}^{(\Im)} \quad (30)$$

where $B_m(f_k)$ are B-spline basis functions evaluated at frequency f_k , K_j and K_{jl} are the numbers of basis functions for the j -th diagonal and (j, l) off-diagonal components, respectively, and $w_{j,m}^{(\delta)}$, $w_{jl,m}^{(\Re)}$, $w_{jl,m}^{(\Im)}$ are the corresponding spline coefficients. When the blocked, coarse-grained likelihood is used, these expansions are evaluated at the coarse-bin centres $\bar{f}_h$, yielding the parameters δ_{jh}^2 and $\theta_{jl}^{(h)}$ that enter Eq. (20).

Knots are placed adaptively, in the spirit of Maturana-Russel and Meyer [7]: for each Cholesky component the empirical periodogram is denoised, the absolute gradient of the result — plus a small uniform floor so that featureless regions still receive knots — is normalised to a probability mass function over frequency, and knots are positioned at equally spaced quantiles of its cumulative distribution. For the off-diagonal entries the real and imaginary parts are treated as separate components, with knots placed independently on $|\Re[\theta_{jl}(f_k)]|$ and $|\Im[\theta_{jl}(f_k)]|$. Knots therefore cluster where the Cholesky component varies rapidly and spread out in smooth regions, with no manual tuning.

Each component j has its own penalty matrix $\mathbf{P}_j$ with entries

$$[\mathbf{P}_j]_{lm} = \int_0^1 B_l''(t) B_m''(t) dt. \quad (31)$$

Unlike the finite-difference approximation $\mathbf{D}_2^\top \mathbf{D}_2$ used in the original P-spline formulation of Eilers and Marx [6], this integral form is exact for non-uniform knot sequences [22]. Note that the integral form of $\mathbf{P}_j$ is the same for every component. The j -subscript reflects only that the knot sequence and basis size K_j are placed adaptively per component, so the resulting $K_j \times K_j$ matrix differs across j .

E. Bayesian Inference

The full inference workflow proceeds in four stages. First, we place hierarchical smoothing priors on the

spline coefficients of every Cholesky component. Second, we combine these priors with the η -tempered, blocked, coarse-grained Whittle likelihood (Eq. 26) using the per-channel factorisation of Eq. (19). Third, we warm-start each MCMC chain from a draw of a stochastic variational approximation to the posterior, to bypass the expensive NUTS adaptation in low-mass-curvature regions. Fourth, we sample the warm-started posterior with NUTS and assess convergence with standard chain diagnostics. Each stage is described below.

We place hierarchical priors on the spline coefficients to enforce smoothness. Each spline component (the diagonal log-variances $\log \delta_j^2$ and the real and imaginary parts of each off-diagonal entry θ_{jl}) is assigned its own pair of precision hyperparameters; with a slight abuse of notation we index a generic such component by j :

$$\mathbf{w}_j | \phi_j \sim \mathcal{N}(\mathbf{0}, (\phi_j \tilde{\mathbf{P}}_j)^{-1}) \quad (32)$$

$$\phi_j | \nu_j \sim \text{Gamma}(\alpha_\phi, \nu_j \beta_\phi) \quad (33)$$

$$\nu_j \sim \text{Gamma}(\alpha_\nu, \beta_\nu) \quad (34)$$

where $\mathbf{w}_j$ is the vector of spline coefficients and ϕ_j, ν_j are precision hyperparameters.² We use $\tilde{\mathbf{P}}_j = \mathbf{P}_j + \epsilon \mathbf{I}$ with $\epsilon = 10^{-6}$ to ensure the prior precision is strictly positive definite. Unless stated otherwise, the results below use $\alpha_\phi = \beta_\phi = 1$ and $\alpha_\nu = \beta_\nu = 1$.

Combining the η -tempered likelihood (Eq. 26) with these priors and the Cholesky factorisation of $\mathbf{S}^{-1}$, the posterior is sampled using the No-U-Turn Sampler [NUTS; 19]. The Cholesky parametrisation guarantees Hermitian positive definiteness of $\hat{\mathbf{S}}(f)$ at every frequency by construction.

a. Computational complexity. The B-spline basis matrices are evaluated once at the start of a run, after the knot locations are fixed, and reused for every gradient evaluation. Each call to $\log \mathcal{L}_\eta$ then costs $\mathcal{O}(p^2 N_c K)$ floating-point operations (using a common basis size $K = K_j = K_{jl}$ for all components): N_c coarse bins, K basis functions per component, and p^2 spline components — the p diagonal log-variances $\log \delta_{jh}^2$ plus the real and imaginary parts of the $p(p-1)/2$ off-diagonal Cholesky entries θ_{jl} . Memory is dominated by the same $p^2 N_c K$ basis-matrix storage. In practice this per-evaluation cost is small compared to the fixed overheads of NUTS warm-up and SVI initialisation, so wall-clock runtime is dominated not by the gradient-evaluation cost but by the number of leapfrog integrator steps required to traverse the posterior. As a rough scale, a single evaluation of $\log \mathcal{L}_\eta$ on the 1-year LISA configuration ($p = 3$, $N_c = 1024$, $K = 100$) takes a fraction of a millisecond on a single CPU core. Empirical end-to-end

² In the closely related univariate constructions of Maturana-Russel and Meyer [7] and Aimen *et al.* [8] this hyperparameter is denoted δ_j . We use ν_j here to avoid collision with the Cholesky diagonal entries δ_{jk}^2 introduced in Section II above.

wall-clock numbers are reported alongside the tables in Section III.

b. Implementation. We implement the model in JAX [23] with NumPyro [24], exploiting automatic differentiation and just-in-time (JIT) compilation for gradient-based sampling. Throughout we use degree-2 B-splines with second-order derivative penalties (the integral form of Eq. (31)). Positive precision parameters ϕ_j and ν_j are sampled on the log scale to keep the sampler away from the boundary at zero, and the frequency grid is standardised to $[0, 1]$ for numerical stability of the B-spline basis evaluation. To initialise the spline parameters we first run stochastic variational inference [SVI; 18] with a low-rank Gaussian guide (rank 16), optimised with Adam [25] at learning rate 10^{-3} for 2000 iterations. Four NUTS chains are then initialised at independent draws from the resulting variational posterior. The NUTS mass matrix is adapted during 2000 warm-up steps, followed by an additional 2000 sampling steps per chain. These sampler settings are used for both the simulation study (Section III A) and the LISA application (Section III B). Only the basis size K , the number of blocks N_b , and the tempering η vary between the two.

c. Convergence diagnostics. We assess convergence by checking that the chains satisfy $\hat{R} < 1.01$ (the rank-normalised Gelman–Rubin potential scale reduction factor [26, 27]), that the bulk and tail effective sample sizes both exceed 400, that the energy-Bayesian fraction of missing information satisfies E-BFMI > 0.3 , and that divergent transitions and maximum tree-depth saturation are absent or negligible. All results in Section III satisfy these criteria unless explicitly noted otherwise.

III. APPLICATION

A. Simulation study

We summarise spectral estimation accuracy with two complementary metrics. The first is the relative integrated squared (Frobenius) error (RISE),

$$\text{RISE} = \frac{\int \|\hat{S}(f) - S(f)\|_F^2 df}{\int \|S(f)\|_F^2 df}, \quad (35)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and the integrals are evaluated numerically using Simpson’s rule over the periodogram frequency grid. This metric measures the bias of the posterior-median estimator $\hat{S}(f)$ relative to the true spectrum $S(f)$. The second metric is the empirical coverage at level α , defined as the fraction of (frequency, matrix-element) pairs for which the true values lie inside the pointwise α credible band, averaged over realisations. This metric measures the calibration of the credible intervals. We use $\alpha = 0.9$ throughout.

We simulate from a three-dimensional VAR(2) process,

$$\mathbf{X}_t = A_1 \mathbf{X}_{t-1} + A_2 \mathbf{X}_{t-2} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad (36)$$

with

$$A_1 = \text{diag}(0.4, 0.3, 0.2), \quad (37)$$

$$A_2 = \begin{bmatrix} -0.2 & 0.5 & 0.0 \\ 0.4 & -0.1 & 0.0 \\ 0.0 & 0.0 & -0.1 \end{bmatrix}, \quad (38)$$

$$\Sigma = \begin{bmatrix} 0.25 & 0.00 & 0.08 \\ 0.00 & 0.25 & 0.08 \\ 0.08 & 0.08 & 0.25 \end{bmatrix}. \quad (39)$$

These coefficient matrices define a stationary process: the companion matrix has spectral radius $0.774 < 1$.

The one-sided theoretical spectral density matrix is

$$\mathbf{S}(f) = \frac{2}{f_s} H(f) \Sigma H(f)^*, \quad \text{with} \quad (40)$$

$$H(f) = \left(I - \sum_{\ell=1}^2 A_\ell e^{-i2\pi f \ell / f_s} \right)^{-1}. \quad (41)$$

We generate 500 independent realisations of length $n = 16,384$ (with a 512-point burn-in) at sampling frequency $f_s = 1$. Each realisation is fit with $K = 10$ basis functions for each Cholesky component, while sweeping $N_b \in \{1, 2, 4, 8, 16\}$ and $N_h \in \{\text{off}, 2, 4, 8, 16, 32\}$. Safe-Bayes tempering (Section II C) is left at $\eta = 1$ throughout, with $N_h = \text{off}$ denoting no coarse-graining. A rectangular window (ENBW = 1) is applied to each block before computing the DFT. All other settings follow Section II E.

Figure 2 shows the full 3×3 spectral matrix recovered for a representative $n = 16,384$ realisation, with the NUTS and SVI posteriors overlaid. Both samplers recover the diagonal PSDs and off-diagonal cross-spectra with substantially smoother posterior medians than the raw periodogram, and the two posterior medians track each other closely (RISE = 0.114 for NUTS versus 0.102 for SVI). The SVI credible bands, however, are narrower than those of NUTS: for this realisation the empirical 90% coverage drops from 88% under NUTS to 83% under SVI. This is the well-known under-dispersion of mean-field/low-rank Gaussian variational approximations, and it is why we use SVI only as a fast NUTS initialiser and retain NUTS as the production sampler when calibrated credible bands are required. That said, the SVI fit itself completes in a few seconds and recovers an accurate posterior median, which is attractive when only a point estimate of the spectrum is needed.

Table I reports a representative slice of the (N_b, N_h) grid. Coverage is near-nominal (89.8–90.8%) across every configuration, and the mean RISE varies by less than 3% across the grid. The number of retained frequency bins N_c falls by a factor of up to 32 between the finest ($N_b = 1$, no coarse-graining) and coarsest ($N_b = 16$, $N_h = 2$) configurations, with no measurable loss of accuracy or calibration. MCMC convergence is satisfactory across all cells: $\hat{R} \leq 1.01$ for all parameters, negligible divergent transitions, and E-BFMI > 0.6 .

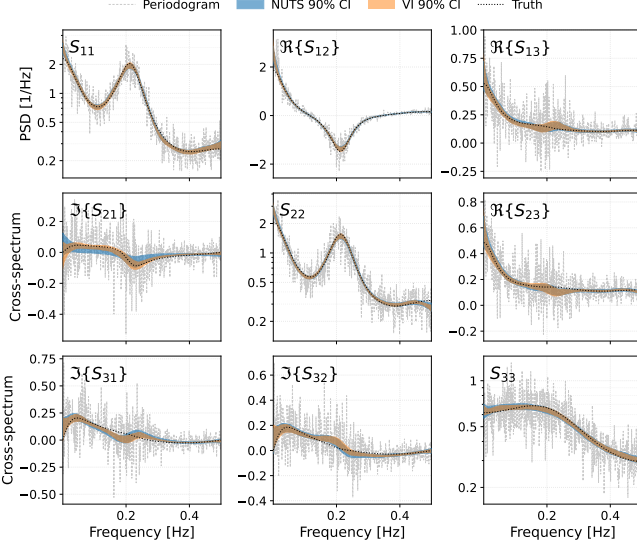


FIG. 2. Representative 3×3 spectral density matrix estimate for a single $n = 16,384$ VAR(2) realisation, fit with $K = 10$, $N_b = 4$, $N_h = 4$ and $\eta = 1$ (NUTS: RISE = 0.114, 90% CI coverage 88%; SVI: RISE = 0.102, 90% CI coverage 83%). Each panel shows the coarse-grained periodogram statistic $\mathbf{Y}_h^{\text{cg}}/(T_b N_b N_h)$ (grey), the theoretical spectrum (dotted black), and posterior medians with 90% credible bands for NUTS (blue) and SVI (orange). Diagonal panels (S_{11} , S_{22} , S_{33}) show the power spectral densities, while the off-diagonal panels show the real and imaginary parts of the cross-spectra on a linear vertical scale (the VAR(2) PSDs span less than one decade, so all panels share a linear scale).

Wall-clock runtime is essentially constant across the entire grid (8.7–9.6 s on a 4-core CPU node) despite the 32-fold variation in N_c . At this problem size ($K = 10$, $p = 3$, $n = 16,384$) the per-gradient cost is small enough that runtime is dominated by fixed NUTS warm-up, JIT compilation, and SVI initialisation overheads rather than by the $\mathcal{O}(p^2 N_c K)$ likelihood evaluation. The benefit of coarse-graining at this scale is therefore primarily statistical (a smaller effective $N_b N_h$ makes the posterior easier to sample) rather than computational. Coarse-graining begins to pay off in wall-clock terms once N_c or K is large enough that the per-gradient cost exceeds the fixed overheads, as in the LISA application below.

B. LISA

We apply the multivariate P-spline model to the three second-generation time-delay interferometry (TDI) Michelson channels X_2 , Y_2 , Z_2 from a publicly released LISA TDI dataset [28, 29]. All channels are TDI 2.0 throughout, and we suppress the generation subscript in what follows, writing XYZ for (X_2, Y_2, Z_2) . The dataset was produced with `lisainstrument` [30] under the LISA Data Challenge (LDC) Spritz noise model [31] at a sampling rate of $f_s = 0.5$ Hz and constant 8.3 s light

travel times. The six inter-spacecraft interferometer (ISI) and test-mass (TM) link measurements were propagated through `pytdi` [32] to produce the TDI 2.0 Michelson combinations used here.

The noise-orthogonal (A, E, T) TDI combinations are obtained from XYZ by the orthonormal rotation $(A, E, T)^\top = M_{\text{AET}}(X, Y, Z)^\top$ with

$$M_{\text{AET}} = \begin{pmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}, \quad (42)$$

which satisfies $M_{\text{AET}}^\top M_{\text{AET}} = I$ and has unit determinant, so the rotation introduces no log-likelihood Jacobian correction [33]. Under the idealised equal-arm, equal-noise assumption the AET basis diagonalises the noise covariance, motivating the diagonal-AET restrictions tested below.

We analyse two noise configurations:

- **noise4a** [28] — symmetric realisation: all six MOSAs are assigned identical Optical Metrology System (OMS; ISI carrier) and test-mass acceleration amplitude spectral densities equal to the LDC baseline values ($2.4 \times 10^{-15} \text{ m s}^{-2}/\sqrt{\text{Hz}}$ and $7.9 \times 10^{-12} \text{ m}/\sqrt{\text{Hz}}$, respectively). In this regime the 3×3 cross-spectral matrix is close to degenerate: $S_{XX} \approx S_{YY} \approx S_{ZZ}$, with off-diagonals equal in magnitude. A closed-form reference is available from the analytic TDI transfer of the link-level ASDs.
- **noise5a** [29] — asymmetric realisation: the six per-MOSA OMS and test-mass ASDs are drawn independently from $\mathcal{U}(0.5, 2.0) \times \text{baseline}$ with a fixed seed. The resulting cross-spectral matrix has distinct diagonals and non-trivial off-diagonal structure, representing the physically relevant regime where mis-modelled cross-spectral correlations would degrade sensitivity to stochastic backgrounds. A closed-form 3×3 reference is built with the `SEGWO` package [34] by feeding the twelve measured per-MOSA ASDs through analytic OMS/TM filter shapes and projecting through PyTDI’s TDI 2.0 Michelson XYZ combinations to obtain the full 3×3 XYZ covariance at every frequency.

We fit each dataset under three competing hypotheses for the 3×3 noise cross-spectrum, designed to test whether the XYZ $\rightarrow$ AET rotation diagonalises the LISA noise covariance to within statistical resolution. Since M_{AET} is orthogonal with unit determinant, fitting in the AET basis is equivalent to fitting in the XYZ basis up to a deterministic relabelling of $\mathbf{S}$. Wherever the AET basis is used ($\mathcal{H}_0$ and $\mathcal{H}_1$), the XYZ $\rightarrow$ AET rotation is applied to the time series first, and all subsequent blocking, averaging, and coarse-graining is performed in the fitting basis. The three hypotheses therefore form a strictly nested hierarchy over the families of representable spectral matrices,

TABLE I. 3D VAR(2) simulation results at $n = 16,384$, $K = 10$, degree-2 splines (mean $\pm$ std over 500 realisations), all runs at $\eta = 1$. Coverage is the fraction of (frequency, matrix-element) pairs inside the pointwise 90% credible interval. RISE is the relative mean squared (Frobenius) error. ESS is the median effective sample size across parameters. $N_h = \text{off}$ denotes no coarse-graining ($N_h = 1$).

N_b	N_h	N_c	Coverage	RISE	ESS	Runtime (s)
1	off	8192	0.900 ± 0.031	0.111 ± 0.004	$21,019 \pm 2726$	9.61 ± 1.76
1	2	4096	0.898 ± 0.031	0.112 ± 0.005	$24,057 \pm 2807$	9.21 ± 1.69
1	4	2048	0.900 ± 0.030	0.112 ± 0.005	$26,352 \pm 3081$	9.03 ± 1.56
1	8	1024	0.904 ± 0.029	0.111 ± 0.005	$26,640 \pm 3303$	8.92 ± 1.65
1	16	512	0.908 ± 0.029	0.110 ± 0.004	$25,480 \pm 3525$	8.81 ± 1.62
1	32	256	0.907 ± 0.030	0.109 ± 0.004	$24,011 \pm 3500$	8.82 ± 1.70
2	off	4096	0.898 ± 0.031	0.112 ± 0.005	$23,938 \pm 2939$	9.11 ± 1.63
2	2	2048	0.901 ± 0.030	0.112 ± 0.005	$26,279 \pm 2979$	9.03 ± 1.72
2	4	1024	0.904 ± 0.030	0.111 ± 0.005	$26,405 \pm 3361$	8.89 ± 1.67
2	8	512	0.907 ± 0.030	0.110 ± 0.004	$25,726 \pm 3461$	8.76 ± 1.75
2	16	256	0.907 ± 0.031	0.109 ± 0.004	$23,997 \pm 3787$	8.81 ± 1.69
4	off	2048	0.901 ± 0.030	0.112 ± 0.004	$26,053 \pm 2906$	9.05 ± 1.71
4	2	1024	0.904 ± 0.029	0.111 ± 0.005	$26,695 \pm 3221$	8.84 ± 1.64
4	4	512	0.907 ± 0.030	0.110 ± 0.004	$25,753 \pm 3636$	8.75 ± 1.72
4	8	256	0.907 ± 0.030	0.109 ± 0.004	$24,036 \pm 3622$	8.78 ± 1.68
8	off	1024	0.904 ± 0.030	0.112 ± 0.005	$26,336 \pm 3190$	8.89 ± 1.65
8	2	512	0.907 ± 0.030	0.111 ± 0.005	$25,587 \pm 3658$	8.78 ± 1.70
8	4	256	0.907 ± 0.031	0.110 ± 0.004	$24,000 \pm 3787$	8.79 ± 1.70
16	off	512	0.905 ± 0.029	0.111 ± 0.004	$25,504 \pm 3431$	8.74 ± 1.55
16	2	256	0.907 ± 0.030	0.110 ± 0.004	$23,862 \pm 3724$	8.84 ± 1.68

$\mathcal{H}_0 \subset \mathcal{H}_1 \subset \mathcal{H}_2$. The three models also differ in their data object: $\mathcal{H}_0$ uses the three diagonal AET periodograms, whereas $\mathcal{H}_1$ and $\mathcal{H}_2$ use the full 3×3 periodogram matrix (in the AET and XYZ bases respectively):

- $\mathcal{H}_0$ (*diagonal AET*). The cross-spectral matrix is constrained to be *exactly* diagonal in the AET basis. The XYZ time series is first rotated into AET via M_{AET} (Eq. 42), and each of the three AET channels is then fit independently with a univariate ($p = 1$) log-P-spline model on the diagonals $\log \delta_j^2$ ($j = 1, 2, 3$, corresponding to $\log S_{AA}, \log S_{EE}, \log S_{TT}$) with $K = 100$ basis functions each. Only the log-diagonal expansion Eq. (28) is used. The off-diagonal expansions Eqs. (29) and (30) are not invoked. The data for this model are the three *univariate* averaged periodograms — the diagonal entries S_{AA}, S_{EE}, S_{TT} of $\mathbf{Y}_h^{\text{cg}}$ — so the likelihood is a product of three independent Gamma (univariate Whittle) likelihoods, one per AET channel; the off-diagonal periodogram entries never enter the fit. This is precisely the special case of the complex Wishart likelihood with a *diagonal* scale matrix $\mathbf{S}$, which factorises into the three diagonal Gamma terms. The diagonal AET posterior is assembled per-sample as $\text{diag}(S_{AA}, S_{EE}, S_{TT})$ and rotated back to the XYZ basis via $\mathbf{S}_{\text{XYZ}} = M_{\text{AET}}^\top \mathbf{S}_{\text{AET}} M_{\text{AET}}$ for diagnostics. The XYZ off-diagonals carried by $\mathcal{H}_0$ therefore reflect only the deterministic TDI rotation, with no fitted physical cross-power.
- $\mathcal{H}_1$ (*multivariate AET with restricted off-diagonals*). The cross-spectral matrix is *approximately* diago-

nal in the AET basis. Unlike $\mathcal{H}_0$, the data here are the full 3×3 averaged AET periodogram matrix and the likelihood is the complete multivariate complex Wishart — the off-diagonal entries do enter the fit. What is restricted is the off-diagonal *model*, not the data. The full multivariate Cholesky model (Eqs. 28–30) is fit to the AET-rotated time series with $K = 100$ basis functions on the diagonals $\log \delta_j^2$ but only $K_\theta = 2$ basis functions on each off-diagonal component θ_{jl} , forcing any fitted cross-power to be smooth and small. $\mathcal{H}_1$ thus occupies the natural middle ground between the strictly diagonal $\mathcal{H}_0$ (off-diagonal data discarded, no cross-power) and the unrestricted $\mathcal{H}_2$: it sees the off-diagonal data but is biased toward a near-diagonal solution, so it can reveal smooth cross-power when present without introducing spurious structure when it is not. The posterior spectral matrix is rotated back to the XYZ basis for diagnostics.

- $\mathcal{H}_2$ (*full multivariate XYZ*). The cross-spectral matrix has fully populated off-diagonals. The multivariate Cholesky model (Eqs. 28–30) is fit directly to the XYZ time series with $K = 100$ basis functions for *every* Cholesky component, both diagonal and off-diagonal. No rotation is needed since the fit is already in the XYZ basis. This is the unrestricted reference model.

For **noise4a** the AET basis diagonalises the noise exactly, so the strong diagonal restriction imposed by $\mathcal{H}_0$ (and the milder one of $\mathcal{H}_1$) are both expected to be valid descriptions, and all three models should agree

within statistical resolution. For **noise5a** the per-MOSA asymmetry introduces genuine off-diagonal power in the AET basis: $\mathcal{H}_0$ is then mis-specified by construction (no off-diagonal flexibility at all), $\mathcal{H}_1$ can absorb only the small-bandwidth component of the off-diagonals through its $K_\theta = 2$ basis functions, and $\mathcal{H}_2$ remains unrestricted. The expected fit-quality ordering is therefore $\text{RISE}(\mathcal{H}_0) > \text{RISE}(\mathcal{H}_1) > \text{RISE}(\mathcal{H}_2)$ for **noise5a** and $\text{RISE}(\mathcal{H}_0) \approx \text{RISE}(\mathcal{H}_1) \approx \text{RISE}(\mathcal{H}_2)$ for **noise4a**.

For all three models, knots are placed by the adaptive quantile rule described in Section II so that more knots fall near the steep low-frequency rise. We choose N_b to give blocks of order one week in length, and a fixed safe-Bayes tempering $\eta = 0.5$ (justified by the sensitivity sweep in Appendix C). Each block is tapered with a Tukey window (taper fraction 0.1, ENBW ≈ 1.04) before computing the DFT to suppress spectral leakage. The three TDI transfer-function nulls at 0.030, 0.060, and 0.090 Hz are excised with half-width 1 mHz prior to fitting. All other settings follow Section II E.

In addition to the RISE introduced in Section III A, we report the median relative width of the 90% credible band on the diagonal PSDs,

$$\Delta_{\text{PSD}} = \text{median}_f \frac{\hat{S}^{95\%}(f) - \hat{S}^{5\%}(f)}{\hat{S}^{50\%}(f)}, \quad (43)$$

computed in the XYZ basis for all three models. This complements the RISE (a bias metric) with a summary of posterior precision. Table II reports RISE, Δ_{PSD} , and wall-clock duration for the three models, two datasets, and three observation durations. All eighteen LISA runs satisfy the convergence diagnostics of Section II E: $\hat{R} < 1.01$ for all parameters, bulk and tail ESS exceeding 400, E-BFMI > 0.3 , and negligible divergent transitions with no tree-depth saturation. The results split sharply along the AET-diagonalisation hypothesis.

Symmetric noise (noise4a). All three models recover the analytic spectral matrix at the half-percent level ($\text{RISE} \approx 4.4 - 5.3 \times 10^{-3}$) across all durations and are statistically indistinguishable from one another. This is the expected behaviour when the AET basis truly diagonalises the noise: $\mathcal{H}_0$ already captures the cross-spectral structure because the only off-diagonal power present in the XYZ basis is the deterministic geometric coupling recovered by the $M_{\text{AET}}^\top \cdot M_{\text{AET}}$ rotation, and the additional off-diagonal flexibility of $\mathcal{H}_1$ and $\mathcal{H}_2$ has nothing to fit. Two conclusions follow: (i) for equal-arm LISA noise the multivariate analysis is not required, and three independent univariate fits suffice, and (ii) the unrestricted $\mathcal{H}_2$ model does not lose accuracy on data that does not need its flexibility, confirming that $\mathcal{H}_2$ remains a safe default when the diagonal-AET assumption cannot be guaranteed in advance.

Asymmetric noise (noise5a). The three models separate by orders of magnitude in the order predicted by their off-diagonal flexibility, $\text{RISE}(\mathcal{H}_0) > \text{RISE}(\mathcal{H}_1) > \text{RISE}(\mathcal{H}_2)$. $\mathcal{H}_0$ stalls at $\text{RISE} \approx 3.3 \times 10^{-2}$ at all durations — a floor set by the AET off-diagonal power it

is structurally unable to represent. $\mathcal{H}_1$ improves on this by roughly 50% ($\text{RISE} \approx 1.8 - 2.2 \times 10^{-2}$) by allowing two off-diagonal basis functions, but is still limited by its narrow θ_{jk} basis. $\mathcal{H}_2$ reaches $\text{RISE} \approx 9 - 11 \times 10^{-4}$, about $20\times$ smaller than $\mathcal{H}_1$ and $30 - 40\times$ smaller than $\mathcal{H}_0$, and continues to improve with duration. This monotone nested ordering is the central diagnostic of the LISA study: when the per-MOSA noise levels are asymmetric (the physically realistic regime) the AET rotation does *not* diagonalise the cross-spectral matrix, and a multivariate model that estimates the full Cholesky factor is required.

Posterior uncertainty contracts as expected. For the multivariate models, the median relative CI width on the diagonal PSDs falls from $\approx 4.7\%$ at 1 month to $\approx 1.2\%$ at 1 year, a reduction of roughly $4\times$, consistent with the $\sqrt{T}$ scaling expected for variance-dominated Whittle inference, and is essentially identical between $\mathcal{H}_1$ and $\mathcal{H}_2$ and across datasets, confirming that the per-channel resolution is set by the data and not by the choice of basis. The univariate $\mathcal{H}_0$ contracts at the same $\sqrt{T}$ rate but with consistently narrower bands ($\approx 3.4\%$ at 1 month, $\approx 0.9\%$ at 1 year), reflecting the smaller per-channel parameter count of the $p = 1$ model when off-diagonals are removed from the inference. For **noise5a** this tighter $\mathcal{H}_0$ posterior is a misleading indicator of fit quality — the bands do not include the truth because the model is structurally unable to represent the AET off-diagonals, as the order-of-magnitude RISE gap to $\mathcal{H}_1$ and $\mathcal{H}_2$ makes clear.

Figure 3 overlays the three durations for both datasets under the $\mathcal{H}_2$ model. The diagonal PSDs are recovered with sub-percent bias and CI widths that contract with $\sqrt{T}$, while the off-diagonal coherences (the cross-channel quantities most directly tied to stochastic-background and parameter-estimation pipelines [35]) are resolved at the few-percent level by 6 months and tightened by a further factor of ~ 1.7 at 1 year. A localised feature visible in all three off-diagonal panels of Fig. 3 is a cluster of small-amplitude wiggles in the posterior-median coherence near the TDI transfer-function null at ≈ 0.06 Hz (and, more weakly, near 0.03 and 0.09 Hz). These arise because the diagonal PSDs $S_{ii}(f)$ drop by several orders of magnitude across each null, and the P-spline basis — which is smooth on a log scale but agnostic to the location and shape of the dips — has to absorb this dynamic range from the data alone. Small residual misfits at the bottoms of the dips are amplified in the coherence, producing the visible ripple. Incorporating the analytic TDI transfer functions directly into the prior mean (rather than only excising a 1 mHz window around each null) may be a natural way to suppress this artefact and is left for future work.

IV. DISCUSSION

We have presented a multivariate P-spline framework for estimating the cross-spectral density matrix of sta-

TABLE II. LISA TDI XYZ results at fixed safe-Bayes tempering $\eta = 0.5$ for the symmetric (**noise4a**) and asymmetric (**noise5a**) datasets, fit with the diagonal-AET null model ($\mathcal{H}_0$, off-diagonal Cholesky coefficients fixed to zero), the AET-rotated multivariate model with restricted off-diagonal basis ($\mathcal{H}_1$, $K_\theta = 2$), and the unrestricted multivariate XYZ model ($\mathcal{H}_2$), at three observation durations. N_b is the number of Wishart time blocks (chosen to give blocks of approximately seven days in length) and N_c is set to 1024. RISE is the relative mean squared (Frobenius) error of the posterior-median spectral matrix against the analytic reference (closed-form equal-link LDC for **noise4a**, SEGWO with measured per-MOSA ASDs for **noise5a**), computed in the XYZ basis for all three models (the AET-basis models are first rotated back to XYZ via M_{AET}). Δ_{PSD} is the median over frequencies of the pointwise relative width $[\hat{S}^{95\%}(f) - \hat{S}^{5\%}(f)]/\hat{S}^{50\%}(f)$ of the 90% credible band on the diagonal PSDs (also in the XYZ basis). Runtime is wall-clock seconds for one full SVI+NUTS pipeline run on a single 4-core CPU node; for $\mathcal{H}_0$ this is the total over the three independent univariate channel fits.

Noise	Duration	N_b	$\mathcal{H}_0$			$\mathcal{H}_1$			$\mathcal{H}_2$		
			RISE	Δ_{PSD}	t (s)	RISE	Δ_{PSD}	t (s)	RISE	Δ_{PSD}	t (s)
4a	1 mo	4	4.3×10^{-3}	3.4%	16	4.5×10^{-3}	4.8%	117	4.7×10^{-3}	4.6%	119
	6 mo	25	4.7×10^{-3}	1.3%	13	4.7×10^{-3}	1.9%	224	5.3×10^{-3}	1.9%	181
	1 yr	52	4.4×10^{-3}	0.8%	244	4.4×10^{-3}	1.2%	632	5.0×10^{-3}	1.2%	853
5a	1 mo	4	3.3×10^{-2}	3.5%	15	2.2×10^{-2}	4.9%	119	1.1×10^{-3}	4.5%	177
	6 mo	25	3.3×10^{-2}	1.4%	13	1.8×10^{-2}	1.9%	218	9.8×10^{-4}	1.9%	245
	1 yr	52	3.3×10^{-2}	1.0%	191	2.2×10^{-2}	1.3%	728	8.6×10^{-4}	1.3%	1089

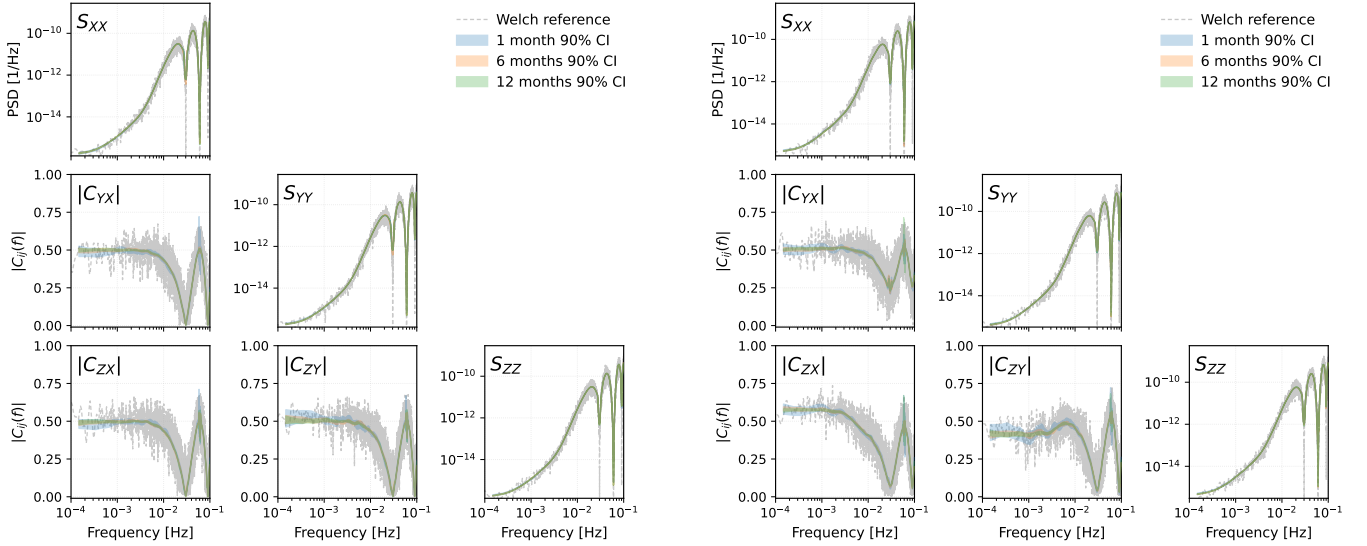


FIG. 3. Posterior summaries of the LISA TDI XYZ spectral matrix from the unrestricted $\mathcal{H}_2$ model for the symmetric **noise4a** (left) and asymmetric **noise5a** (right) datasets at safe-Bayes tempering $\eta = 0.5$, shown at three observation durations: 1 month (blue), 6 months (orange), and 12 months (green). Diagonal panels show the posterior 90% credible bands for the diagonal PSDs S_{XX}, S_{YY}, S_{ZZ} . Lower-triangle panels show the coherence $|C_{ij}(f)|$ (Eq. 4) for each pair. Welch reference (grey) is overplotted.

tionary vector time series. The Cholesky parametrisation of $\mathbf{S}^{-1}$ guarantees Hermitian positive definiteness at every frequency and factorises the likelihood into p independent per-channel regressions that can be sampled in parallel. A blocked, coarse-grained Whittle likelihood combined with safe-Bayes η -tempering yields near-nominal 90% credible-interval coverage (≈ 0.90) across a wide (N_b, N_h) grid on the 3D VAR(2) benchmark, while reducing the number of inference-time frequency bins by up to a factor of 32.

On the synthetic 3D VAR(2) benchmark the untempered estimator ($\eta = 1$) achieves coverage ≈ 0.90 at every (N_b, N_h) configuration tested. On the LISA TDI

data the same un-tempered estimator over-concentrates. A fixed safe-Bayes tempering $\eta = 0.5$ recovers credible bands that contract with $\sqrt{T}$ and remain well-behaved across all three observation durations (Appendix C).

a. η -tempering and debiased Whittle. The η -sweep of Appendix C shows that credible-band width is a strong, monotone function of η at fixed (N_b, N_h, K) , but does not yet identify a principled rule for choosing η across regimes. Mapping the optimal η as a function of N_b, N_h, K , and the underlying spectral curvature is the most pressing methodological question raised by this work. A complementary direction is to replace η -tempering with a debiased Whittle likelihood [36], which

corrects the finite-sample bias of the periodogram analytically rather than down-weighting the entire likelihood.

b. Informative priors. The hierarchical Gaussian smoothing prior used here is near-uninformative: each weight has marginal variance set by a high-level hyperparameter (ϕ_j, ν_j) with no scale imposed by the data ahead of inference. In practice this places no bound on where the spline curves can sit, and most of NUTS warm-up is spent locating a region of weight space consistent with even a coarse Welch estimate. A more informative prior, centred on a fast initial estimate of $\log \delta_j^2(f)$ and the off-diagonal Cholesky terms (e.g. from a Welch-and-smooth pre-pass), with ϕ_j controlling allowed deviations from that estimate, may (i) make the smoothing hyperparameters physically meaningful as fractional excursion bounds, (ii) shorten warm-up substantially, and (iii) reduce the prior-data contention that drives over-coverage at small η in the appendix sweep.

c. Non-stationarity. The Whittle likelihood assumes wide-sense stationarity. Both **noise4a** and **noise5a** are stationary by construction: realisations of an LDC noise model with constant per-MOSA amplitude spectral densities and constant 8.3s light travel times. The real LISA noise environment is non-stationary on multiple timescales — constellation breathing modulates the arm lengths over the year-long orbital period, and per-link noise levels drift on timescales from hours (thermal cycling, glitches) to months (component aging). A natural and tractable extension is to assume *piecewise* stationarity: partition the mission timeline into segments short enough that within-segment stationarity is a good approximation but long enough that the Whittle approximation remains accurate, and either fit each segment independently or couple consecutive segments through a smoothness prior on the frequency-dependent Cholesky parameters in the time direction. This would extend the present model to a time-frequency spectral matrix $\mathbf{S}(t, f)$ at the cost of one additional smoothing dimension.

d. Improved variational families. As reported in Section III A, the current low-rank ($r = 16$) multivariate Gaussian SVI guide recovers an accurate posterior median but produces narrower credible bands than NUTS, with a corresponding drop in empirical coverage. NUTS is therefore retained as the production sampler, and SVI is used only as an initialiser. Because the model is implemented in NumPyro, the variational family is a swappable component: replacing the Gaussian guide with a normalising flow (e.g. a neural spline flow or a masked autoregressive / inverse autoregressive flow) is essentially a one-line change. A systematic study of which variational families recover calibrated PSD posteriors — so that SVI could serve as a stand-alone approximation at a fraction of the NUTS wall-clock cost — is left for future work.

e. Other extensions. The within-bin constant- $\mathbf{S}$ approximation underlying coarse-graining (Eq. 13) can bias estimates near sharp spectral features. Adaptive bin widths that narrow around lines and transfer-function nulls are a natural extension. Hybrid para-

metric/nonparametric models that embed known instrumental features within the P-spline envelope are another route to improving sharp-feature accuracy without sacrificing the smoothness elsewhere.

Data and Software Availability

The software developed for this project is open-source and publicly available from the GitHub repository <https://github.com/nz-gravity/LogPSplinePSD>, which contains all source code, example scripts, and configuration files needed to reproduce the results. Installation instructions and dependency information (tested on Python ≥ 3.10) are provided in the repository **README**. The in-text code links (🔗) are pinned to the tagged release **v0.1.0** of the package, so that the referenced line numbers remain valid as the codebase evolves. The software is released under the MIT License, permitting free use, modification, and distribution. The LISA time series data are available at [28, 29]. Scripts for the LISA analysis are available from the GitHub repository https://github.com/nz-gravity/lisa_multivar_logpsplines_study.git.

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Appendix A: Notation

Table III lists the symbols used throughout the paper, grouped by the stage of the model in which they appear.

Symbol	Meaning
<i>Time series & sampling</i>	
T, Δ_t, f_s	Total observation duration, sampling interval, sampling frequency; $T = n\Delta_t$, $\Delta_t = 1/f_s$.
f_{Ny}	Nyquist frequency, $f_{Ny} = f_s/2$.
n, p	Number of time samples per channel, number of channels.
Δ_f, f_k	Frequency resolution $\Delta_f = 1/T$; k th DFT frequency $f_k = k\Delta_f$.
N	Number of positive Fourier frequencies, $N = \lfloor n/2 \rfloor$.
<i>Blocking & coarse-graining</i>	
N_b, T_b	Number of non-overlapping time blocks; block duration $T_b = T/N_b$ (n/N_b samples per block).
$N_h, J_h, \bar{f}_h$	Number of fine frequencies aggregated per coarse bin; J_h the h th bin (consecutive subset of fine frequencies), with midpoint $\bar{f}_h$.
N_c	Number of coarse-grain bins, $N_c = n/(2N_bN_h)$.
<i>Spectral density</i>	
$\mathbf{Z}$	Observed p -channel time series, $\mathbf{Z} \in \mathbb{R}^{n \times p}$.
$\mathbf{d}(f_k)$	DFT vector, $\mathbf{d}(f_k) \in \mathbb{C}^p$.
$\Gamma(\tau), \gamma_{lm}(\tau)$	Autocovariance matrix at integer lag τ and its (l, m) entry.
$\mathbf{S}(f_k)$	$p \times p$ Hermitian positive-definite spectral density matrix; Fourier transform of Γ .
<i>Periodogram & Wishart statistics</i>	
$\mathbf{I}^{(i)}(f_k), \bar{\mathbf{I}}(f_k)$	Block- i periodogram, $\mathbf{I}^{(i)} = \mathbf{d}^{(i)}\mathbf{d}^{(i)*} \sim \mathcal{CW}_p(T_b\mathbf{S}, 1)$; block-averaged periodogram $\bar{\mathbf{I}} = N_b^{-1} \sum_i \mathbf{I}^{(i)}$.
$\mathbf{Y}(f_k)$	Summed periodogram / Wishart statistic, $\mathbf{Y} = N_b\bar{\mathbf{I}} \sim \mathcal{CW}_p(T_b\mathbf{S}, N_b)$.
$\mathbf{Y}_h^{\text{cg}}$	Coarse-grained statistic, $\mathbf{Y}_h^{\text{cg}} = \sum_{f_k \in J_h} \mathbf{Y}(f_k) \sim \mathcal{CW}_p(T_b\mathbf{S}(\bar{f}_h), N_bN_h)$.
$\lambda_\nu^{(k)}, \mathbf{v}_\nu^{(k)}, \mathbf{u}_\nu^{(k)}$	ν th eigenvalue/unit eigenvector of $\mathbf{Y}(f_k)$; scaled eigenvector $\mathbf{u}_\nu^{(k)} = \sqrt{\lambda_\nu^{(k)}} \mathbf{v}_\nu^{(k)}$.
<i>Cholesky parametrisation</i>	
$\mathbf{T}_k, \mathbf{D}_k$	Factors in $\mathbf{S}(f_k)^{-1} = \mathbf{T}_k^* \mathbf{D}_k^{-1} \mathbf{T}_k$: $\mathbf{T}_k$ unit lower-triangular, $\mathbf{D}_k$ diagonal with entries $\delta_{1k}^2, \dots, \delta_{pk}^2$.
$\theta_{jl}^{(k)}, \delta_{jk}^2$	Off-diagonal entry (j, l) of $\mathbf{T}_k$ ($j > l$) and j th diagonal entry of $\mathbf{D}_k$.
θ_j, δ_j	Parameter vectors for channel j : collect $\theta_{jl}^{(h)}$ ($l < j$) and δ_{jh} across the coarse bins $h = 1, \dots, N_c$.
<i>P-spline model</i>	
$K, B_m(f_k)$	Number of B-spline basis functions (model hyperparameter, distinct from frequency index k); m th basis evaluated at f_k .
K_j, K_{jl}	Basis sizes for the j th diagonal and (j, l) off-diagonal components.
$\mathbf{w}_j^{(\delta)}, \mathbf{w}_{jl}^{(\Re)}, \mathbf{w}_{jl}^{(\Im)}$	Spline coefficient vectors for $\log \delta_{jk}^2, \Re[\theta_{jl}], \Im[\theta_{jl}]$.
$\mathbf{P}_j, \phi_j$	Penalty matrix $[\mathbf{P}_j]_{lm} = \int_0^1 B_l'' B_m'' dt$ and smoothing precision hyperparameter.

TABLE III. Notation used throughout the paper. Symbols are grouped by the stage of the model they belong to.

Appendix B: Bivariate VAR(2) benchmark

This appendix benchmarks the proposed estimator against the bivariate Bayesian baselines available in the literature on a controlled VAR(2) problem with closed-form ground truth. Following Liu *et al.* [20], we simulate 500 independent realisations of a bivariate VAR(2) time series at three sample sizes $n = 256, 512, 1024$ (refer to Liu *et al.* [Section 4.2, 20] for the definitions of the VAR(2) models), and the theoretical spectral density matrix provides exact ground truth. The spectral densities are estimated using the multivariate P-spline method

with $K = 16$ basis functions and degree-2 splines ($d = 2$) for each spectral component. We compare to the variational Bayes (VB) and variational nonparametric correction (VNPC) methods of Liu *et al.* [20] using the same L_2 error criterion. The L_2 error is the un-normalised integrated Frobenius error of the posterior-median spectral matrix, without dividing by $\int \|S(f)\|_F df$. We use it here for direct comparison with the VB and VNPC results, which are reported in this metric.

Table IV summarises the L_2 errors across the 500 independent realisations. The multivariate P-spline estimator is competitive with both comparators at every sam-

TABLE IV. Bivariate VAR(2) benchmark using the L_2 error metric from Liu *et al.* [20]. Entries are mean $\pm$ standard deviation over 500 independent realisations. Smaller values are better.

n	P-spline	VB	VNPC
256	0.095 ± 0.022	0.121 ± 0.029	0.129 ± 0.033
512	0.072 ± 0.018	0.091 ± 0.027	0.103 ± 0.025
1024	0.055 ± 0.013	0.066 ± 0.016	0.082 ± 0.017

ple size and attains the lowest mean error in all three settings. The mean L_2 error decreases from 0.095 at $n = 256$ to 0.072 at $n = 512$ and 0.055 at $n = 1024$, compared with 0.121, 0.091, and 0.066 for VB and 0.129, 0.103, and 0.082 for VNPC. This is the main message we want from the bivariate benchmark: the proposed multivariate P-spline method achieves accuracy of the same order as established alternatives, while in these simulations being modestly more accurate on average.

Appendix C: η -tempering sensitivity on noise4a

To support the choice of tempering parameter for the LISA application, we re-ran the `noise4a` dataset at

$\eta \in \{0.01, 0.03, 0.1, 0.25, 0.5, 0.75, 1\}$ for each of the three observation durations (1, 6, and 12 months) used in Section III B, holding all other settings fixed. Figure 4 summarises the result.

Two features emerge from Figure 4. First, the matrix RISE is essentially flat across the entire sweep: it varies by less than $\pm 5\%$ about its mean at every duration, confirming that the posterior-median point estimate is nearly insensitive to η . Second, credible-band width contracts monotonically with η , by roughly an order of magnitude from $\eta = 0.01$ to $\eta = 1$, with the steepest contraction at the longest duration—consistent with the per-bin Fisher information scaling as $N_b N_h$ (Eq. (27)).

The vertical dotted line marks the choice $\eta = 0.5$ used in Section III B. This value gives a practical compromise: credible bands are meaningfully wide at all three durations while the point estimate remains accurate throughout. Reaching nominal coverage would require yet smaller η or a more flexible model (e.g. adaptive coarse-graining near the TDI nulls). We leave this to future work.

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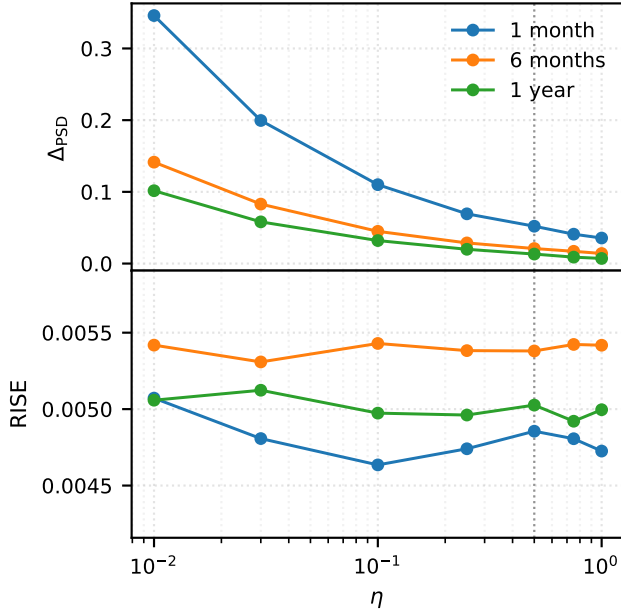


FIG. 4. Safe-Bayes η -sensitivity on the symmetric LISA **noise4a** dataset at three observation durations (1 month, 6 months, 1 year). Top: median pointwise relative width of the 90% credible band on the diagonal PSDs. Bottom: matrix RISE (narrow y -range—RISE varies by $\lesssim 5\%$ across the full sweep). The vertical dotted line marks $\eta = 0.5$, the value used in Section III B.

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